**Procedure: Training a Basic Federated Learning Model Using TensorFlow Federated**

**What is Federated Learning?**

Federated Learning is a machine learning technique in which multiple devices train a shared model using their local data without directly sharing the raw data with a central server.

**Example:** Two mobile phones have different datasets. Both train a model on their own data and send model updates to a central server. The server combines these updates to improve the shared model.

**Step-by-Step Procedure**

1. **Install the Required Libraries:** Install Python and the required libraries, such as TensorFlow and TensorFlow Federated.
2. **Prepare the Dataset:** Select a small dataset, such as MNIST, which contains handwritten digit images from 0 to 9.
3. **Simulate Multiple Devices:** Divide the dataset into smaller parts to represent data stored on different devices, such as Device 1, Device 2, and Device 3.
4. **Create a Machine Learning Model:** Build a simple neural network using TensorFlow to recognize handwritten digits.
5. **Initialize the Federated Learning Process:** Use TensorFlow Federated to initialize the global model and configure the federated training process.
6. **Train the Model Locally:** Each simulated device trains the model using its own local dataset without sending the raw data to the server.
7. **Send Model Updates:** Each device sends its model updates to the central server instead of sharing its original dataset.
8. **Aggregate Model Updates:** The server combines the updates from multiple devices to improve the global model. This can be done using Federated Averaging (FedAvg).
9. **Repeat Training Rounds:** The updated global model is distributed back to the devices, and local training is repeated for several rounds.
10. **Evaluate the Model:** Test the final model using a separate test dataset and calculate its accuracy.
11. **Record the Results:** Record the model's accuracy, number of training rounds, and observations about how the model learns from multiple devices.

**Expected Result**

The federated learning model learns from data distributed across multiple simulated devices without directly collecting their raw training data on the central server.

**Conclusion:** Federated Learning enables multiple devices to collaboratively train a shared machine learning model while keeping their raw training data local. However, model updates can still reveal information, so additional privacy protections may be required.

# ORAL

Federated Learning – Viva Questions and Answers

1. What is federated learning, and how does it differ from centralized training?

Answer: Federated Learning (FL) is a machine learning technique in which multiple devices train a shared model without sending their raw data to a central server. In centralized training, data from different sources is collected on a central server for training.

2. Which data and which messages move between the clients and the server in one round?

Answer: The server sends the global model to the clients. Each client trains the model using its local data and sends model updates, such as weights or gradients, back to the server. The raw training data stays on the client.

3. Explain the Federated Averaging algorithm and why the average is weighted by the number of samples.

Answer: Federated Averaging (FedAvg) combines model updates from multiple clients to create an improved global model. The average is weighted by the number of samples so that clients with more training data contribute proportionally more to the result.

4. What is TensorFlow Federated, and how are devices simulated in it?

Answer: TensorFlow Federated (TFF) is an open-source framework for developing federated learning systems. Devices are simulated by dividing a dataset into separate client datasets and treating each dataset as an individual client during training.

5. What is non-IID data, and why does it slow down or reduce the accuracy of federated training?

Answer: Non-IID data means that different clients have different data distributions. For example, one device may contain mostly images of digits 1 and 2, while another contains mostly digits 8 and 9. This can make it harder for the global model to learn effectively from all clients.

6. What is client drift, and how can it be reduced?

Answer: Client drift occurs when local models move in different directions because clients have different datasets. It can be reduced by limiting local training steps, adjusting learning rates, and using methods such as FedProx.

7. Why can federated learning still leak private information even though raw data never leaves the device?

Answer: Model updates can sometimes reveal information about the original training data. Attackers may analyze gradients or model parameters to infer sensitive information.

8. How do secure aggregation and differential privacy improve the privacy of federated learning?

Answer: Secure aggregation allows the server to obtain combined client updates without seeing each client's individual update. Differential privacy adds carefully controlled noise to reduce the possibility of identifying information about individual users.

9. What are the main communication and system challenges of federated learning on real devices?

Answer: The main challenges include slow internet connections, high communication costs, limited device memory and processing power, battery consumption, and devices having different amounts of data.

10. Name some real-world applications of federated learning and explain why FL suits them.

Answer: Applications include:

- Keyboard prediction: Learns typing patterns while keeping personal messages on devices.
- Healthcare: Allows hospitals to collaborate on model training without directly sharing patient records.
- Industrial IoT: Helps machines learn from sensor data while keeping operational data local.

- Mobile devices: Improves personalized services while reducing the need to collect raw user data centrally.

Federated learning is suitable for these applications because it enables collaborative model training while helping organizations and users maintain greater control over their data.